

CURRENT ELECTRICITY

Important formulas and concepts directly from NCERT-

❖ Ohm's Law-

$$V \propto I$$
$$V = IR$$

➤ $R = \rho l/A$

➤ $E = j\rho$ (E = electric field

$j = \sigma E$ j = current density

ρ = resistivity and σ = conductivity)

❖ Drift velocity-

$$v_d = -eE\tau/m$$

➤ Conductivity,

$$\sigma = m/ne^2\tau$$

➤ Mobility,

$$\mu = |v_d| / E = e\tau/m$$

❖ Limitations of ohm's law -

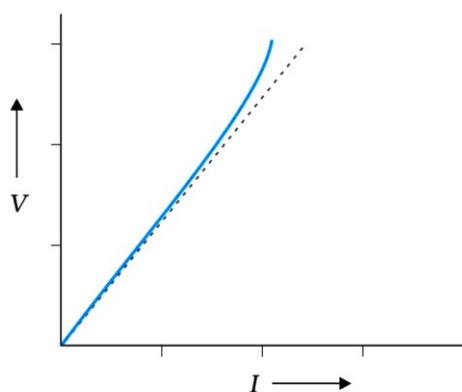


FIGURE 3.5 The dashed line represents the linear Ohm's law. The solid line is the voltage V versus current I for a good conductor.

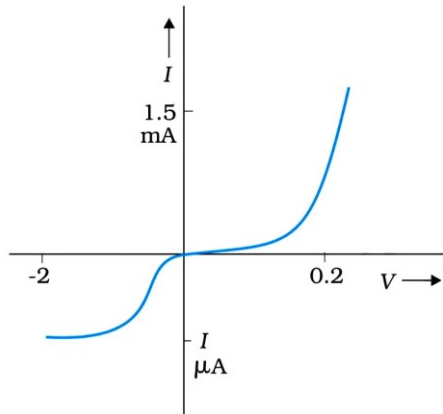


FIGURE 3.6 Characteristic curve of a diode. Note the different scales for negative and positive values of the voltage and current.

2)

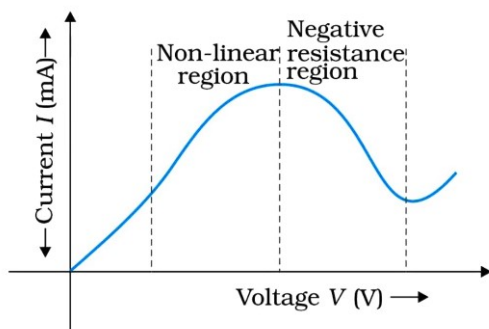


FIGURE 3.7 Variation of current versus voltage for GaAs.

3)

❖ Temperature dependence of Resistivity-

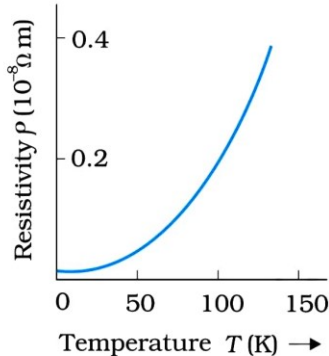


FIGURE 3.8 Resistivity ρ_T of copper as a function of temperature T .

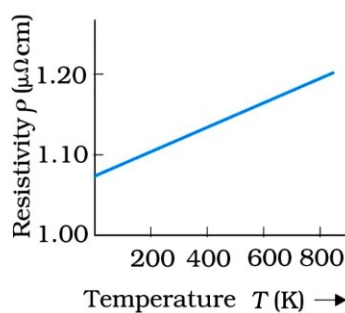


FIGURE 3.9 Resistivity ρ_T of nichrome as a function of absolute temperature T .

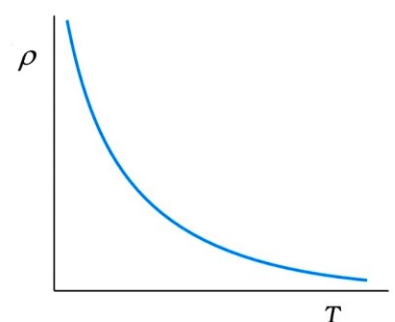


FIGURE 3.10 Temperature dependence of resistivity for a typical semiconductor.

- **Nichrome (an alloy of nickel, iron and chromium), manganin and constantan are used in wire bound standard resistors as their R change very little with T .**

- For conductors, ρ increases with temp.
- For insulators and semiconductors, ρ decreases with temp.

❖ Power-

$$P = I^2 R = V^2 / R$$

❖ Cell-

- Potential difference,

$$V = \varepsilon - Ir$$

(V = potential difference
 I = current
 ε = emf of cell
 r = internal resistance of cell)

- Current,

$$I = \varepsilon / (R+r)$$

Maximum current for $R=0$, which is ε/r .

❖ Series and parallel combination of cells-

- In series,

$$\varepsilon_{eq} = \varepsilon_1 + \varepsilon_2$$

$$r_{eq} = r_1 + r_2$$

- In parallel,

$$\varepsilon_{eq} = \varepsilon_1 r_2 + \varepsilon_2 r_1 / r_1 + r_2$$

$$r_{eq} = r_1 + r_2$$

❖ Wheatstone bridge-



➤ **Balanced condition for the galvanometer to give zero or null deflection,**

$$R_1/R_2 = R_3/R_4$$

SUMMARY

1. *Current* through a given area of a conductor is the net charge passing per unit time through the area.
2. To maintain a steady current, we must have a closed circuit in which an external agency moves electric charge from lower to higher potential energy. The work done per unit charge by the source in taking the charge from lower to higher potential energy (i.e., from one terminal of the source to the other) is called the electromotive force, or *emf*, of the source. Note that the emf is not a force; it is the voltage difference between the two terminals of a source in open circuit.
3. *Ohm's law*: The electric current I flowing through a substance is proportional to the voltage V across its ends, i.e., $V \propto I$ or $V = RI$, where R is called the *resistance* of the substance. The unit of resistance is ohm: $1\Omega = 1\text{ V A}^{-1}$.

4. The *resistance* R of a conductor depends on its length l and cross-sectional area A through the relation,

$$R = \frac{\rho l}{A}$$

where ρ , called *resistivity* is a property of the material and depends on temperature and pressure.

5. *Electrical resistivity* of substances varies over a very wide range. Metals have low resistivity, in the range of $10^{-8}\Omega\text{ m}$ to $10^{-6}\Omega\text{ m}$. Insulators like glass and rubber have 10^{22} to 10^{24} times greater resistivity. Semiconductors like Si and Ge lie roughly in the middle range of resistivity on a logarithmic scale.
6. In most substances, the carriers of current are electrons; in some cases, for example, ionic crystals and electrolytic liquids, positive and negative ions carry the electric current.
7. *Current density* $\mathbf{j}$ gives the amount of charge flowing per second per unit area normal to the flow,

$$\mathbf{j} = nq \mathbf{v}_d$$

where n is the number density (number per unit volume) of charge carriers each of charge q , and $\mathbf{v}_d$ is the *drift velocity* of the charge carriers. For electrons $q = -e$. If $\mathbf{j}$ is normal to a cross-sectional area $\mathbf{A}$ and is constant over the area, the magnitude of the current I through the area is $nev_d A$.

8. Using $E = V/l$, $I = nev_d A$, and Ohm's law, one obtains

$$\frac{eE}{m} = \rho \frac{ne^2}{m} v_d$$

The proportionality between the *force* eE on the electrons in a metal due to the external field E and the drift velocity v_d (not acceleration) can be understood, if we assume that the electrons suffer collisions with ions in the metal, which deflect them randomly. If such collisions occur on an average at a time interval τ ,

$$v_d = a\tau = eE\tau/m$$

where a is the acceleration of the electron. This gives

$$\rho = \frac{m}{ne^2\tau}$$

9. In the temperature range in which resistivity increases linearly with temperature, the *temperature coefficient of resistivity* α is defined as the fractional increase in resistivity per unit increase in temperature.
10. Ohm's law is obeyed by many substances, but it is not a fundamental law of nature. It fails if

- (a) V depends on I non-linearly.
- (b) the relation between V and I depends on the sign of V for the same absolute value of V .
- (c) The relation between V and I is non-unique.

An example of (a) is when ρ increases with I (even if temperature is kept fixed). A rectifier combines features (a) and (b). GaAs shows the feature (c).

11. When a source of emf $\mathcal{E}$ is connected to an external resistance R , the voltage V_{ext} across R is given by

$$V_{ext} = IR = \frac{\mathcal{E}}{R + r} R$$

where r is the *internal resistance* of the source.

12. *Kirchhoff's Rules* –

- (a) *Junction Rule*: At any junction of circuit elements, the sum of currents entering the junction must equal the sum of currents leaving it.
- (b) *Loop Rule*: The algebraic sum of changes in potential around any closed loop must be zero.

13. The *Wheatstone bridge* is an arrangement of four resistances – R_1 , R_2 , R_3 , R_4 as shown in the text. The null-point condition is given by

$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

using which the value of one resistance can be determined, knowing the other three resistances.

POINTS TO PONDER

1. Current is a scalar although we represent current with an arrow. Currents do not obey the law of vector addition. That current is a scalar also follows from its definition. The current I through an area of cross-section is given by the scalar product of two vectors:

$$I = \mathbf{j} \cdot \Delta \mathbf{S}$$

where $\mathbf{j}$ and $\Delta \mathbf{S}$ are vectors.

2. Refer to V - I curves of a resistor and a diode as drawn in the text. A resistor obeys Ohm's law while a diode does not. The assertion that $V = IR$ is a statement of Ohm's law is not true. This equation defines resistance and it may be applied to all conducting devices whether they obey Ohm's law or not. The Ohm's law asserts that the plot of I versus V is linear i.e., R is independent of V .

Equation $\mathbf{E} = \rho \mathbf{j}$ leads to another statement of *Ohm's law*, i.e., a conducting material obeys Ohm's law when the resistivity of the material does not depend on the magnitude and direction of applied electric field.

3. Homogeneous conductors like silver or semiconductors like pure germanium or germanium containing impurities obey Ohm's law within some range of electric field values. If the field becomes too strong, there are departures from Ohm's law in all cases.
4. Motion of conduction electrons in electric field $\mathbf{E}$ is the sum of (i) motion due to random collisions and (ii) that due to $\mathbf{E}$. The motion due to random collisions averages to zero and does not contribute to v_d (Chapter 10, Textbook of Class XI). v_d , thus is only due to applied electric field on the electron.
5. The relation $\mathbf{j} = \rho \mathbf{v}$ should be applied to each type of charge carriers separately. In a conducting wire, the total current and charge density arises from both positive and negative charges:

$$\mathbf{j} = \rho_+ \mathbf{v}_+ + \rho_- \mathbf{v}_-$$

$$\rho = \rho_+ + \rho_-$$

Now in a neutral wire carrying electric current,

$$\rho_+ = -\rho_-$$

Further, $v_+ \sim 0$ which gives

$$\rho = 0$$

$$\mathbf{j} = \rho_- \mathbf{v}$$

Thus, the relation $\mathbf{j} = \rho \mathbf{v}$ does not apply to the total current charge density.

6. Kirchhoff's junction rule is based on conservation of charge and the outgoing currents add up and are equal to incoming current at a junction. Bending or reorienting the wire does not change the validity of Kirchhoff's junction rule.